FREE GUIDE

How to store lyophilised peptides in the lab

Freeze-dried (lyophilised) peptides are among the more stable forms laboratory material comes in — until they are mishandled. This guide covers the conditions that keep a sample stable and the everyday habits that quietly degrade it.

At a glance

State	Short term	Longer term	Notes
Lyophilised powder, sealed	2–8 °C (refrigerator)	-20 °C or colder (freezer)	Keep sealed, dry and dark. Follow the supplier's documentation where it differs.
Lyophilised powder, opened	2–8 °C, tightly resealed	Aliquot, then -20 °C	Every opening admits moisture. Minimise the number of times a vial is opened.
In solution	2–8 °C	Aliquot and freeze; avoid repeated thawing	Solutions are far less stable than powder. Stability depends on the sequence, solvent and pH.

Always defer to the batch documentation. Storage conditions on the certificate or product sheet are specific to that material and take precedence over general guidance, including this guide.

Before and after reconstitution

Once a powder is dissolved, the rules change. Water allows reactions that are effectively frozen in the dry state — hydrolysis, oxidation and, for some sequences, aggregation. A powder that is stable for a long period in a freezer may be stable for a much shorter period in solution.

Use the solvent specified in the supplier's documentation, prepare only what is needed for the work planned, and record the date of preparation on the container. Where a solution must be kept, divide it into single-use aliquots so that the remaining stock is never thawed and refrozen.

For laboratory and research use only. Not for human or veterinary use. Not medical advice.

Pepco Lab · SEE BEE DEE LIMITED · Registered in England and Wales no. 17072052 · pepcolab.com

WHAT DEGRADES A SAMPLE

The four enemies of stability

1. **Heat**
   Rates of chemical degradation rise with temperature. Keep material cold, keep time at room temperature short, and do not store vials in a fridge door, where temperature swings each time it opens.

2. **Moisture**
   Lyophilised powder readily absorbs water from the air. Let a cold vial reach room temperature before opening it — opening it cold causes condensation inside. Reseal promptly; a desiccant in the secondary container helps.

3. **Light**
   Some residues (for example tryptophan, tyrosine and those in copper complexes) are sensitive to light. Store in the original amber or opaque packaging, or in a closed box.

4. **Freeze–thaw cycles**
   Each cycle stresses a solution: ice crystals form, solutes concentrate, pH can shift. Aliquoting once is far better than thawing a single stock repeatedly.

Oxygen, and the sequences that mind it most

Methionine, cysteine and tryptophan residues oxidise more readily than others. For material containing them, minimise headspace, keep containers sealed, and consider an inert-gas overlay where your laboratory practice allows it.

Signs a sample may have changed

What you see	What it may mean
Powder has collapsed, become sticky or glassy	Moisture uptake. The material may still be usable, but should be treated with caution.
Discolouration	Possible oxidation or contamination.
Cloudiness or particles in solution	Incomplete dissolution, aggregation or contamination.
Results drift between experiments	Degradation in a stock that has been stored too long or thawed too often.

For laboratory and research use only. Not for human or veterinary use. Not medical advice.

Pepco Lab · SEE BEE DEE LIMITED · Registered in England and Wales no. 17072052 · pepcolab.com

BENCH-READY

Labelling, logging and handling

Good storage is mostly good habits. These keep material traceable and make problems visible early.

On arrival

- [ ] Check the package arrived cool and intact, and note the condition.
- [ ] Match the lot number on each vial to its certificate of analysis, and file the certificate.
- [ ] Move material to the correct storage temperature without delay.

Every container

- [ ] Compound, lot number, and concentration (for solutions).
- [ ] Date received, date opened, and date prepared (for solutions).
- [ ] Storage condition and the initials of whoever prepared it.

Every use

- [ ] Allow cold vials to reach room temperature before opening.
- [ ] Use clean, dry tools; never return unused material to the stock vial.
- [ ] Reseal and return to storage promptly; log the use against the lot.

Periodically

- [ ] Check freezer and refrigerator temperatures, ideally with a logged thermometer.
- [ ] Review stock against preparation dates; discard what is past its documented stability.
- [ ] Keep certificates with the inventory record so any result can be traced to its lot.

Why this matters for your results. A certificate describes material at the moment it was tested. Storage is what decides whether that description is still true by the time the material is used. Traceable lots plus consistent storage are what make results comparable over time.

Pepco Lab dispatches from cold storage in insulated packaging. Questions about storing a specific material: hello@pepcolab.com

For laboratory and research use only. Not for human or veterinary use. Not medical advice.

Pepco Lab · SEE BEE DEE LIMITED · Registered in England and Wales no. 17072052 · pepcolab.com